Yangxin Wu

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Education

Harbin Institute of Technology, Shenzhen — Shenzhen, China
Bachelor's Degree, Computer Science and Technology — 09/2023 - Present

Research Interests

- **AI agents for scientific R&D**: agentic AI for autonomous scientific discovery; long-horizon research agents that propose hypotheses, design and run experiments, use tools and memory, and iterate from feedback to accelerate scientific research.
- **Scalable post-training and RL**: verifier-/environment-driven learning frameworks, automated evaluation, online distillation, on policy distillation, and self-improvement to extend LLM reasoning and agentic capabilities beyond static human demonstrations with minimal human-prior assumptions.

Research Experience

Solving PDE-constrained optimization problems via operator learning — 03/2026 - Present
Supervisor: Tailin Wu, Assistant Professor — Westlake University

- Developed an neural operator surrogate for PDE-constrained optimization, approximating the control/parameter-to-state map to reduce repeated high-fidelity forward and adjoint PDE solves.
- Adapted Evolutionary Ensemble of Agents scaffold to co-evolve executable frameworks and agent guidance states; coordinated multiple coding agents to optimize loss formulations, training strategies, and optimization pipelines.

Experiment-grounded reinforcement learning environments for tokamak plasma control — 03/2026 - 05/2026
Supervisor: Tailin Wu, Assistant Professor — Westlake University

- Developed TokamaCon, a GPU-vectorized parallel RL environment suite for experiment-grounded plasma control, with standardized tasks, partial-observation interfaces, safety-aware termination, reproducible metrics, and PPO-based reference baselines.
- Validated the environments against EAST reference simulators and demonstrated scalable large-batch rollout performance, enabling reproducible evaluation of learned controllers under distribution shift and hard safety constraints.
- The paper "TokamaCon: An Experiment-Grounded Parallel Environment Suite for Tokamak Plasma Control" was submitted to NeurIPS 2026.

Higher-order graph neural networks as neural PDE solver — 03/2025 - 05/2025
Supervisor: Xiucheng Li, Assistant Professor — Harbin Institute of Technology, Shenzhen

- Proposed a higher-order graph neural network framework based on discrete and finite element exterior calculus for solving PDEs, achieving state-of-the-art performance on boundary value problems in electromagnetism.
- The paper "Boundary-Value PDEs Meet Higher-Order Differential Topology-aware GNNs" was published at NeurIPS 2025 (Spotlight Accepted).

Low-resource speech translation with large-scale LM and ASR models — 01/2025 - 04/2025
Supervisor: Kehai Chen, Professor — Harbin Institute of Technology, Shenzhen

- Integrated open-source large language models with sequence-to-sequence automatic speech recognition models to train speech translation systems for each language direction.
- Achieved 1st place in the Unconstrained End-to-End systems for three directions (Bn-En, Hi-En, Ta-En) in the IWSLT 2025 Evaluation Campaign — Indic Track. The system description paper was published in the IWSLT 2025 Proceedings.

Technical Background

- Object-oriented and functional programming with Python, especially PyTorch